

☒ Already in the live page☐ Missing from the live page — add this☐ Fires, but not a customer toggle (show as info)

Alerts and Alarms

Notifications for your solar system: energy usage, battery state of charge, weather forecasts, and capacity utilization.

Solar system alerts

1. Get energy usage alerts

IN CURRENT UI

Choose alert frequency

Daily energy usage alerts ☐Weekly energy usage alerts ☐

Daily / weekly checkboxes already exist.
Times stay hardcoded (daily 10:00, weekly
Monday 10:00).

Which days get the 10:00 daily report

MISSING FROM CURRENT UI

Backend now sends the daily report **every calendar day at 10:00** whose weekday is in `data.daily_energy_usage_days` (Mon=0 ... Sun=6). Default is all 7 days. Uncheck Sat/Sun to skip weekends. The report covers **yesterday**. Live page has no day chips, so customers cannot exclude weekends from Alerts.

Mon

Tue

Wed

Thu

Fri

Sat

Sun

2. Send battery SOC alerts by email

IN CURRENT UI☐

Maps to `battery_soc_config.email_enabled` and
`data.daily_battery_soc_alerts`.

Also notify the mobile app (push)

IN CURRENT UI☒

Maps to `battery_soc_config.push_enabled` only. Do not turn this off when they only uncheck email.

Scheduled updates

TIMES IN CURRENT UI**WEEKDAYS MISSING**

Send a status at each time (Africa/Lagos), on the selected weekdays. Granularity 15 minutes. Live page only stores the clock time and always uses every day (0, 1, 2, 3, 4, 5, 6). POST `days_of_week` per time — Mon=0 ... Sun=6.

6:00 AM ×

Mon

Tue

Wed

Thu

Fri

Sat

Sun

5:00 PM ×

Mon

Tue

Wed

Thu

Fri

Sat

Sun

18:00



Add time

When battery charge crosses

IN CURRENT UI

Alert once when SOC hits a level you set. Will not repeat until SOC recovers by about 3%.

≤ 20% ×

Drops to / below (≤) ▼

30

%

Add threshold

3. Night battery over-usage

Follows SOC email ☐

MISSING FROM CURRENT UI

Read-only. Do not POST `solar_power_demand_alerts`. Backend copies it from the SOC *email* switch. Fires 18:00–07:00 when load is higher than the battery can sustain until sunrise. Once per night. Email + push.

With email off (this screenshot), this night alert stays off after the next save that sends `daily_battery_soc_alerts: false`. Show the state so the customer understands it follows “Send battery SOC alerts by email”.

4. Daily Unfavourable Weather Alerts

IN CURRENT UI



06:00 Lagos, hardcoded. On/off only. Email + push when today's forecast is poor for solar.

5. When solar capacity utilization exceeds 90 % of installed capacity

IN CURRENT UI



Any time of day. At most one email/push per hour while still over the %.

6. Daily solar generation recap (app)

MISSING FROM CURRENT UI

NO TOGGLE

Always on. 19:00 Lagos push + in-app for every solar branch with today's generation > 0. Type `daily_solar_usage.alert_setting_key` is null — there is nothing to POST. Not the same as item 1 (10:00 usage report email). Show this as copy so customers know the 7pm app message is expected.

7. Solar panel soiling

MISSING FROM CURRENT UI

NO TOGGLE

No Alerts-page toggle and no standalone push. If last week's yield is low vs weather, a **banner on the Monday weekly solar report**

email is added. Type solar_soiling.

Save Updates

Correctly hidden on solar-only

Do not render these when the nested config is null: energy_usage_config (needs EMS meters), diesel_entry_reminder_config (needs a generator), power-quality / generator rows (voltage, PF, load balance, CO₂, operating hours).

What the live page missed vs this template

Gap	What to do	Payload
Weekdays on the 10:00 daily report	Mon–Sun chips under Daily energy usage. Default every day. Uncheck Sat/Sun to skip weekends.	<code>data.daily_energy_usage_days</code> e.g. <code>"0,1,2,3,4,5,6"</code>
Weekdays on SOC times	Per-time Mon–Sun chips. Default every day.	<code>battery_soc_config.schedule_times[].days_of_week</code> e.g. <code>"0,1,2,3,4,5,6"</code>
Night battery over-usage	Show as read-only, follows SOC email. Do not POST the boolean.	GET <code>data.solar_power_demand_alerts</code> (read-only)
19:00 solar	Info row only. No	Nothing to save. Type <code>daily_solar_usage</code>

generation push	checkbox.	
Soiling	Info row only. Banner on weekly email.	Nothing to save. Type <code>solar_soiling</code>

Endpoint

`GET /api/v1/alerts_data/<branch_id>/` → 200 · `POST` same URL → 202. JWT required.

GET (solar-only, matches this screen)

POST body (what Save should send)

When they fire

```
{
  "status": true,
  "message": "Successful",
  "data": {
    "id": 41,
    "branch": 180,
    "operating_time_alerted_at": "2020-01-01T00:00:00",
    "power_factor_alerted_at": "2020-01-01T00:00:00",
    "energy_usage_alerted_at": "2020-01-01T00:00:00",
    "load_balance_alerted_at": "2020-01-01T00:00:00",
    "emitted_co2_alerted_at": "2020-01-01T00:00:00",
    "frequency_alerted_at": "2020-01-01T00:00:00",
    "baseline_alerted_at": "2020-01-01T00:00:00",
    "voltage_alerted_at": "2020-01-01T00:00:00",
    "load_alerted_at": "2020-01-01T00:00:00",
    "power_factor_alerts": false,
    "max_power_factor": 0,
    "min_power_factor": 0,
    "baseline_alerts": false,
    "energy_usage_alerts": false,
    "energy_usage_max": 0,
    "load_balance_alerts": false,
    "frequency_alerts": false,
    "frequency_normal": 50,
    "frequency_precision": 0,
    "voltage_alerts": false,
    "max_voltage": 0,
    "min_voltage": 0,
    "emitted_co2_alerts": false,
    "set_co2_alerts": false,
    "set_co2_value": 0,
    "operating_time_alerts": false,
    "operation_start_time": null,
    "operation_end_time": null,
    "load_alerts": false,
    "load_threshold_value": 0,
    "changeover_lag_alerts": false,
    "generator_maintenance_alert": false,
    "daily_energy_usage_alerts": false,
    "daily_energy_usage_days": "0,1,2,3,4,5,6",
    "weekly_energy_usage_alerts": false,
    "daily_battery_soc_alerts": false,
    "daily_unfavorable_weather_alerts": true,
    "solar_power_demand_alerts": false,
  }
}
```

```
"solar_power_demand_alerted_at": "2020-01-01T00:00:00",
"solar_capacity_utilization_alerts": true,
"solar_capacity_utilization_threshold_pct": 90,
"solar_capacity_utilization_alerted_at": "2020-01-01T00:00:00",
"diesel_entry_reminder_alerts": false,
"diesel_entry_reminder_alerted_on": null
},
"generator_data": [],
"battery_soc_config": {
  "id": 12,
  "branch_id": 180,
  "branch_name": "BR Space",
  "notification_type": "daily_battery_soc",
  "category": "battery",
  "is_enabled": true,
  "push_enabled": true,
  "email_enabled": false,
  "schedule_times": [
    {
      "id": 1,
      "time": "06:00",
      "days_of_week": "0,1,2,3,4,5,6",
      "last_sent_on": null
    },
    {
      "id": 2,
      "time": "17:00",
      "days_of_week": "0,1,2,3,4,5,6",
      "last_sent_on": null
    }
  ],
  "thresholds": [
    {
      "id": 5,
      "metric": "battery_soc",
      "operator": "lte",
      "value": 20
    }
  ],
  "updated_at": "2026-09-24T17:28:00"
},
"energy_usage_config": null,
"diesel_entry_reminder_config": null,
"catalog": {
  "categories": [
    {
      "code": "energy",
      "label": "Energy"
    },
    {
      "code": "battery",
      "label": "Battery"
    },
    {
      "code": "solar",
      "label": "Solar"
    },
    {
      "code": "diesel",
      "label": "Diesel"
    },
    {
      "code": "power_quality",
      "label": "Power quality"
    }
  ]
}
```

```
    },
    {
      "code": "operations",
      "label": "Operations"
    },
    {
      "code": "environment",
      "label": "Environment"
    }
  ],
  "types": [
    {
      "code": "daily_energy_usage",
      "label": "Daily Energy Usage",
      "category": "energy",
      "configurable": false,
      "surfaces": [
        "push",
        "email"
      ],
      "alert_setting_key": "daily_energy_usage_alerts"
    },
    {
      "code": "daily_solar_usage",
      "label": "Daily Solar Usage",
      "category": "solar",
      "configurable": false,
      "surfaces": [
        "push"
      ],
      "alert_setting_key": null
    },
    {
      "code": "daily_battery_soc",
      "label": "Battery State of Charge",
      "category": "battery",
      "configurable": true,
      "supports_times": true,
      "supports_thresholds": true,
      "eligibility": "solar_battery",
      "surfaces": [
        "push",
        "email",
        "alerts_page"
      ],
      "alert_setting_key": "daily_battery_soc_alerts",
      "alerts_data_field": "battery_soc_config"
    },
    {
      "code": "over_usage_solar_power_demand",
      "label": "Battery Over-Usage",
      "category": "battery",
      "configurable": false,
      "surfaces": [
        "push",
        "email"
      ],
      "alert_setting_key": "solar_power_demand_alerts"
    },
    {
      "code": "over_usage_solar_capacity",
      "label": "Solar Capacity Over-Usage",
      "category": "solar",
      "configurable": false,
```

```
    "eligibility": "solar_inverter",
    "surfaces": [
      "push",
      "email",
      "alerts_page"
    ],
    "alert_setting_key": "solar_capacity_utilization_alerts"
  },
  {
    "code": "energy_usage_target",
    "label": "Energy Usage Target",
    "category": "energy",
    "configurable": true,
    "eligibility": "ems_meters",
    "surfaces": [
      "push",
      "email",
      "alerts_page"
    ],
    "alert_setting_key": "energy_usage_alerts",
    "alerts_data_field": "energy_usage_config"
  },
  {
    "code": "diesel_entry_reminder",
    "label": "Diesel Entry Reminder",
    "category": "diesel",
    "configurable": true,
    "eligibility": "generator",
    "surfaces": [
      "push",
      "email",
      "alerts_page"
    ],
    "alert_setting_key": "diesel_entry_reminder_alerts",
    "alerts_data_field": "diesel_entry_reminder_config"
  },
  {
    "code": "daily_unfavorable_weather",
    "label": "Unfavourable Solar Weather",
    "category": "solar",
    "configurable": false,
    "eligibility": "solar_station",
    "surfaces": [
      "push",
      "email",
      "alerts_page"
    ],
    "alert_setting_key": "daily_unfavorable_weather_alerts"
  },
  {
    "code": "weekly_energy_usage",
    "label": "Weekly Energy Usage Report",
    "category": "energy",
    "configurable": false,
    "surfaces": [
      "email",
      "alerts_page"
    ],
    "alert_setting_key": "weekly_energy_usage_alerts"
  },
  {
    "code": "solar_soiling",
    "label": "Solar Panel Soiling",
    "category": "solar",
```

```
        "configurable": false,
        "eligibility": "solar_station",
        "surfaces": [
            "email"
        ],
        "alert_setting_key": null,
        "delivery": "weekly_report_banner"
    }
],
"alert_settings": [
    {
        "key": "daily_energy_usage_alerts",
        "label": "Daily energy usage report",
        "category": "energy",
        "group": "standard",
        "fires": true
    },
    {
        "key": "daily_energy_usage_days",
        "label": "Daily report weekdays (Mon=0...Sun=6)",
        "category": "energy",
        "group": "standard",
        "fires": true
    },
    {
        "key": "weekly_energy_usage_alerts",
        "label": "Weekly energy usage report",
        "category": "energy",
        "group": "standard",
        "fires": true
    },
    {
        "key": "daily_unfavorable_weather_alerts",
        "label": "Daily unfavourable weather",
        "category": "solar",
        "group": "standard",
        "fires": true
    },
    {
        "key": "solar_capacity_utilization_alerts",
        "label": "Solar capacity utilization",
        "category": "solar",
        "group": "standard",
        "fires": true
    },
    {
        "key": "daily_battery_soc_alerts",
        "label": "Battery SOC (email)",
        "category": "battery",
        "group": "battery",
        "fires": true
    },
    {
        "key": "solar_power_demand_alerts",
        "label": "Night battery over-usage",
        "category": "battery",
        "group": "battery",
        "fires": true
    }
]
}
}
```

